`elegant-polimi-thesis` documentation

This chapter is the result of the documentation generated from the source code. Since it's not strictly needed to use the template – most of the functions are straightforward – it's quite barebones.

This document contains them all, even those that are not exposed to the user, which are marked with "_" before.

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Theses templates

polimi-thesis

The main thesis formatting function.

Parameters

```
polimi-thesis(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str arr,
  academic-year: str,
  tutor: str,
  cycle: str,
  chair: str,
  student-id: str,
  course: str,
  frontispiece: "phd" "deib-phd" "cs-eng-master" "classical-master",
  custom-logo: image,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Advisor"`

coadvisor `str` or `arr`

Coadvisor(s) of the thesis.

Default: `"Coadvisor"`

academic-year `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

tutor `str`

Tutor of the thesis.

Default: `"Tutor"`

cycle `str`

Cycle of the thesis.

Default: `"XXV"`

chair `str`

Chair of the thesis.

Default: `"Chair"`

student-id `str`

Student ID.

Default: `"00000000"`

course `str`

Student course.

Default: `"Course Engineering"`

frontispiece `"phd"` or `"deib-phd"` or `"cs-eng-master"` or `"classical-master"`

Frontispiece of the thesis. Can be either: `phd`, `deib-phd`, `cs-eng-master` or `classical-master`.

Default: `"phd"`

custom-logo `image`

Custom logo of the thesis.

Default: `none`

polimi-article-format-thesis

The thesis article format styling function.

Parameters

```
polimi-article-format-thesis(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str array,
  academic-year: str,
  student-id: str,
  course: str,
  abstract: content,
  keywords: str,
  logo: path,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Prof. Name Surname"`

coadvisor `str` or `array`

Coadvisor(s) of the thesis.

Default: `"Prof. Name Surname"`

academic-year `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

student-id `str`

Student ID.

Default: `"00000000"`

course `str`

Student course.

Default: `"Xxxxxxxxxxxx Engineering - Ingegneria Xxxxxxxxxxxx"`

abstract `content`

Abstract.

Default: `[]`

keywords `str`

Keywords, that appear below the abstract (and in the PDF metadata).

Default: `"word, word, word"`

logo `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

polimi-executive-summary

The executive summary styling function.

Parameters

```
polimi-executive-summary(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str array,
  academic-year: str,
  course: str,
  logo: path,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Prof. Name Surname"`

coadvisor `str` or `array`

Coadvisor(s) of the thesis.

Default: `"Prof. Name Surname"`

academic-year `str`

Academic year of the thesis. If empty, defaults to "2025-2026".

Default: `""`

course `str`

Student course.

Default: `"Xxxxxxxxxxxx Engineering - Ingegneria Xxxxxxxxxxxx"`

logo `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

_blank-toc

Helper function to manually add blank space above an outline entry (similar to LaTeX's `\addtocontents{toc}{\vspace{1em}}`).

Parameters

`_blank-toc(space: length) -> content`

space `length`

Vertical space to add before the next outline entry.

Default: `1em`

frontmatter

Frontmatter section. Similar to LaTeX's `\frontmatter`, it is meant to be only used in the thesis. It sets the page numbering to `"i"` and `numbering: none` for headings.

Parameters

```
frontmatter(body) -> content
```

acknowledgements

Acknowledgements section. It sets `numbering: none` for headings.

Parameters

```
acknowledgements(body) -> content
```

mainmatter

Mainmatter section. Similar to LaTeX's `\mainmatter`, it is meant to be only used in the thesis. It sets to page numbering to `"1"`, heading numbering to `"1.1" + h()` and resets the page counter.

Parameters

```
mainmatter(body) -> content
```

appendix

Appendix section. Similar to LaTeX's `\appendix`. It sets heading numbering to `"A.1"` and resets their counter.

Parameters

```
appendix(body) -> content
```

backmatter

Backmatter section. Similar to LaTeX's `\backmatter`, it is meant to be only used in the thesis. It sets heading numbering to `none`.

Parameters

```
backmatter(body) -> content
```

_lists

Lists figure to make the list of tables, list of figures to appear in the table of contents.

Parameters

```
_lists() -> content
```

_lists-entries-style

Internal helper function to create the custom lists of figures and table.

Parameters

```
_lists-entries-style(
  outline-entry: outline-entry,
  kind: function
) -> content
```

outline-entry `outline-entry`

Outline entry to edit.

kind `function`

The kind of the outline entry element (image or table)

nomenclature

Displays a simple nomenclature with keys and values.

```
#nomenclature((
    "Polimi": "Politecnico di Milano",
    "CdL": "Corso di Laurea",
    "CCS": "Consigli di Corsi di Studio",
    "CFU": "Crediti Formativi
Universitari",
))
```

Nomenclature

POLIMI	Politecnico di Milano
CDL	Corso di Laurea
CCS	Consigli di Corsi di Studio
CFU	Crediti Formativi Universitari

Parameters

```
nomenclature(
  dict: dictionary,
  indented: bool
) -> content
```

dict `dictionary`

Dictionary that hold keys and values.

indented `bool`

Whether to indent or not the nomenclature.

Default: `true`

theorems-init

Utility function to initialize the theorem environments[1]:

- theorem
- proposition
- lemma
- remark

Parameters

```
theorems-init(body) -> content
```

subfigure

Creates a new subfigure with the given arguments and an optional label. From hallon package.

Parameters

```
subfigure(
  ..args,
  outlined: bool,
  label: str,
  body
) -> content
```

outlined `bool`

Whether to outline this subfigure or not.

Default: `false`

label `str`

Unique label to reference this subfigure

Default: `none`

toc `content`

Custom-built `outline()`.

list-of-figures `content`

List of figures. Similar to LaTeX's `\listoffigures`.

[1]Provided by great-theorems package.

list-of-tables `content`

List of tables. Similar to LaTeX's `\listoftables`.

theorem `content`

Theorem block.

proposition `content`

Proposition block.

lemma `content`

Lemma block.

remark `content`

Remark block.

proof `content`

Proof block.

Utilities

_is-page-empty

Check if a page is empty.

Parameters

```
_is-page-empty() -> bool
```

_empty-page

Adds an empty page between an odd page and the next. Used to check when to remove the header and place a raggiera in the bottom left corner.

Parameters

```
_empty-page() -> content
```

_nameref

It displays a reference using section name (instead of numbering). From Hallon 0.1.3

Parameters

```
_nameref(label) -> content
```

_raggiera-image

Inserts a raggiera, given a specified width.

Parameters

```
_raggiera-image(width: length) -> content
```

width `length`

Width of the raggiera.

banner

Draw a banner.

Parameters

```
banner(body: content) -> content
```

body `content`

Body of the banner.

tab-numbering

Alternative numbering: `"1.1." + h-space`.

Parameters

```
tab-numbering(
  h-space: length,
  ..n: arguments
) -> content
```

h-space `length`

Horizontal space.

Default: `0.7em`

..n `arguments`

Numbers.

chapter-numbering

Chapter numbering.

Parameters

```
chapter-numbering(..n: arguments) -> content
```

..n `arguments`

Numbers.

header-number

Numbering used in the theses header.

Parameters

```
header-number(..n: arguments) -> content
```

..n `arguments`

Numbers.

_spacer

Empty block.

Parameters

```
_spacer(
  width,
  height
) -> content
```

_stroke-no-border

Internal stroke.

Parameters

```
_stroke-no-border(color) -> stroke
```

_poli-header

Custom made header.

Parameters

```
_poli-header(
  text-args: dict,
  self
) -> content
```

text-args `dict`

The text arguments for the title.

Default: `(size: 15pt, weight: "regular", fill: rgb("#5e5e5e"))`

_poli-bg-split

Apply a fill colour to the panes.

Parameters

```
_poli-bg-split() -> content
```

_divider-body

New section body, that is title, lettering and chapter number.

Parameters

```
_divider-body(
  title,
  lettering,
  number,
  text-fill
) -> content
```

_divider-bg-args

Style the new section background.

Parameters

```
_divider-bg-args(
  fill,
  stroke
) -> content
```

_basic-subslide-preamble

Standard subslide preamble.

Parameters

```
_basic-subslide-preamble(
  self,
  inset,
  text-args
) -> content
```

_sizes `dict`

LaTeX sizes to match original templates (https://tex.stackexchange.com/questions/24599/what-point-pt-font-size-are-large-etc)

_document-state `state`

The current state (title page, front-,main-,backmatter...)

_document-type `state`

The current type of document.

_localization `dict`

Localization dictionary. It could be deprecated since thesises are written exclusively in English.

bluepoli `color`

Signature PoliMi colour (■), used in headings and labels.

_pane `dict`

The background panes used in some scenarios.

_custom-numbering `numbering`

(1..20) -> (01,02,03,...,20)

Figures styling

_subfigure-caption

It displays the caption of subfigures. Use "(a)" that will be colored based on `colored-headings` parameter.

Parameters

```
_subfigure-caption(
  colored-caption: bool,
  parent: figure,
  it
) -> content
```

colored-caption `bool`

Whether to apply bluepoli on captions

Default: `true`

parent `figure`

Parent figure of this subfigure

Default: `none`

_style-figures

Style-figures handles (optional heading-dependent) numbering of figures and subfigures. Can modify the caption of all figures.

Parameters

```
_style-figures(
  colored-caption: bool,
  heading-levels: int,
  body
) -> none
```

colored-caption `bool`

Whether to apply bluepoli on captions (Figure 1).

Default: `true`

heading-levels `int`

Heading level to account for in the numbering.

Default: `0`

Presentation

poli-slide

Basic slide.

Parameters

```
poli-slide(
  config: dictionary,
  repeat: auto bool,
  setting,
  composer,
  ..args,
  body
) -> content
```

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

repeat `auto` or `bool`

Whether to repeat this slide.

Default: `auto`

title-slide

The title slide. The style can be specified via the corresponding argument. For the background and logos style you must also specify a background and logos.

Parameters

```
title-slide(
  style: "default" "white" "logos" "background",
  background: image,
  logos: array,
  config: dictionary
) -> content
```

style `"default"` or `"white"` or `"logos"` or `"background"`

Style of the title slide.

Default: `"default"`

background `image`

Required background for the “background” style; otherwise ignored.

Default: `none`

logos `array`

Required logos for the “logos” style; otherwise ignored.

Default: `()`

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

make-outline

Generate the outline based on headings.

Parameters

```
make-outline(config: dictionary) -> content
```

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

exact-split-slide

Slide exactly split in 2 or 4. Requires content for each split.

Parameters

```
exact-split-slide(
  splits: arr,
  config: dictionary
) -> content
```

splits `arr`

Content of each split. Usually a title with a small description.

Default: `()`

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

split-slide

Split a slide in two panes – similar to header spacing.

Parameters

```
split-slide(
  left: content,
  right: content,
  config: dictionary
) -> content
```

left `content`

Left split content.

Default: `[]`

right `content`

Right split content.

Default: `[]`

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

plot-slide

Slide with coloured background. Used for plots.

Parameters

```
plot-slide(
  config: dictionary,
  body
) -> content
```

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

image-slide

Slide with image on the left, right or as whole background. Deprecated. Use `background-slide()`.

Parameters

```
image-slide(
  image: image,
  style: "background" "left",
  config: dictionary,
  body
) -> content
```

image `image`

Image for the slide.

Default: none

style `"background"` or `"left"`

Style of the image, that is where it will be placed and sized.

Default: `"background"`

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

background-slide

Slide with image as whole background.

Parameters

```
background-slide(
  image: image,
  config: dictionary
) -> content
```

image `image`

Image for the slide.

Default: none

config `dictionary`

Touying overrides for this slide.

Default: `(:)`

focus-slide

Last slide. Usually it contains “Thanks for listening!” or similar phrases.

Parameters

```
focus-slide(
  lettering: content,
  config,
  body
) -> content
```

lettering `content`

Bottom lettering.

Default: `none`

_new-section-slide

The so-called “Divisorio”, a new section. Invoked with a level 1 heading.

Parameters

```
_new-section-slide(
  config,
  ..args
) -> content
```

_new-subsection-slide

The other so-called “Divisorio”, but for lvl.2 heading and in a light background. Unused.

Parameters

```
_new-subsection-slide(
  config,
  ..args
) -> content
```

polimi-digital-presentation

Main presentation function.

Parameters

```
polimi-digital-presentation(
  divider-style: "dark" "light",
  subtitle-every-style: bool,
  ..args,
  body
) -> content
```

divider-style `"dark"` or `"light"`

The style of the divider.

Default: `"dark"`

subtitle-every-style `bool`

Whether to use the subtitle on all title styles.

Default: `false`